

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 11: Geometric Progression

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The general term

1. 1701 is the n th term of the G.P. 7, 21, 63, ... Find (a) n (b) the sum of these n terms.

ICSE 2025, 3 marks

SOLUTION

(a) $a = 7, r = 3. 7 \times 3^{n-1} = 1701$, so $3^{n-1} = 243 = 3^5$ and $n = 6$.

(b) $S_6 = \frac{a(r^6 - 1)}{r - 1} = \frac{7(729 - 1)}{2} = 7 \times 364$

Answer: (a) $n = 6$ (b) sum = 2548

2 Sum of n terms

2. 15, 30, 60, 120, ... is a G.P. (a) Write its n th term. (b) How many terms add up to 945?

ICSE 2024, 4 marks

SOLUTION

(a) $a = 15$ and $r = 2$, so $T_n = 15 \times 2^{n-1}$.

(b) $S_n = \frac{a(r^n - 1)}{r - 1} = 15(2^n - 1) = 945$

$2^n - 1 = 63$, so $2^n = 64 = 2^6$.

Answer: (a) $T_n = 15 \times 2^{n-1}$ (b) 6 terms

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.